

The Operating System for Electromagnetic Spectrum

Executive Primer

Distribute as Needed

v3.0 | 2026

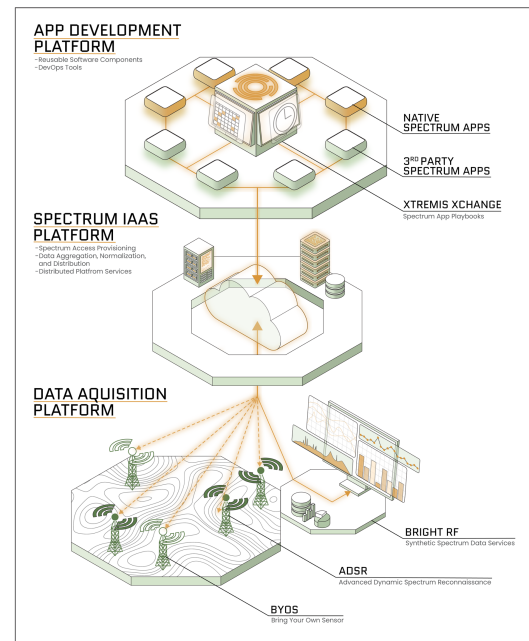
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THE PROBLEM

The electromagnetic spectrum underpins every modern system of consequence: 5G, GPS, radar, satellite communications, electronic warfare, autonomous vehicles. It is also, for almost every organization that depends on it, **functionally invisible**. Governments allocate it on paper. Militaries plan around it weeks in advance. Commercial operators treat it as static input. Nobody can see what is actually happening in the spectrum in real time, at scale, with enough fidelity to act.

This electromagnetic opaqueness converges from three directions.

Spectrum Waste

The U.S. has auctioned over \$230B in spectrum licenses since 1994. The DoD controls roughly a third of usable spectrum below 6 GHz and uses only 20–35% on average. Every 100 MHz of mid-band spectrum released contributes up to \$264B in GDP over 15 years. The waste is staggering—and the root cause is zero real-time visibility into actual usage.

The Drone Threat

Counter-UAS is no longer a niche military problem. Russia produces over 50,000 fiber-optic FPV drones per month—vehicles with zero RF emissions undetectable by conventional intercept. AI-autonomous drones navigate on vision alone. Domestically, unauthorized drone incursions over critical infrastructure are accelerating. Existing C-UAS was designed for RF-emitting threats. That assumption is obsolete. Detection of RF-dark targets requires passive sensing, fused with acoustic, EO/IR, and RF exploitation under a single C2 layer. No fielded system does this.

Electronic Warfare Dominance

Ukraine made this unambiguous. The EM spectrum is the most contested domain in modern conflict: both sides jam GPS, spoof drone links, and hunt emitters as targeting cues. EW capability—the ability to see and act in the spectrum faster than the adversary—is now as decisive as fires. The U.S. military spent decades on kinetic overmatch while underinvesting in spectrum awareness.

The convergence: Spectrum waste, battlefield EM dominance, and the drone threat share a single root cause—the inability to see and act in the electromagnetic spectrum in real time. Xtremis AI exists to solve that problem.

THE COMPANY

Xtremis AI is a Vanderbilt University spinout founded by the team that won two DARPA Grand Challenges in RF spectrum optimization. We built **spectrumOS**: a distributed, AI-driven platform that senses, comprehends, and orchestrates electromagnetic spectrum in real time. We are the only company with a vertically integrated stack—custom sensing hardware, edge AI, cloud orchestration—proven under operational conditions with the U.S. military. Veteran-owned.

\$100M+ DoD contract portfolio | **\$22.5M** booked 2025 | **Profitable** | **Bootstrapped** | **18–24 mo** tech lead

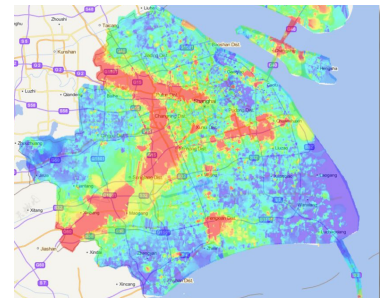
spectrumOS	Distributed OS for real-time spectrum management. Runs at edge and cloud (IL4/IL5). Unifying layer across all capabilities.
SpECTRE	Edge AI for signal detection, classification, anomaly alerting. Zero-shot detection of novel emitters—no prior training data.
VESpER	RF DevOps for rapid CI/CD of spectrum applications. Under contract with Army Research Lab. DC–18 GHz.
Spectrum Exchange	Machine-speed spectrum allocation for competing users and systems. Military deconfliction and commercial trading.
BrightRF	Physics-informed RF propagation modeling and synthetic data generation. Terrain, foliage, weather, hardware aware.
ADSR / RogueRF	Purpose-built sensing hardware: vehicle, manpack, UAS, leave-behind form factors. DC–6 GHz (18 GHz in dev).

OPERATIONAL PROOF POINTS

These systems operate in contested electromagnetic environments along-side maneuver forces. This is not a simulation company.

Spectrum Situational Awareness

ADSR sensors and spectrumOS are deployed with U.S. DoD units across Romania, Iraq, Syria, Sweden, and Germany. The mission: real-time RF situational awareness across the operational environment. Detect, classify, geolocate, and track every emitter—friendly, hostile, neutral. Measured performance: 150 km radiosonde detection, 38.5 km ADS-B tracking, sub-degree angle-of-arrival accuracy. Six international deployments since 2023.



Spectrum activity heatmap

Counter-UAS & RF - Dark Drone Detection

Passive bistatic radar using wideband SDR architecture to detect RF-dark drones via ambient signal reflections (FM, LTE, 5G, satellite illuminators of opportunity). Combined with SkyGuard (drone link interception) and Red Hunter (emitter tracking), spectrumOS becomes a multi-sensor fusion C2 layer. Devil's Den Proving Ground is the development and validation testbed. The C-UAS market is bifurcating: legacy vendors sell RF intercept for yesterday's threat. Fiber-optic and AI-autonomous drones require a fundamentally different architecture built on passive EM observation.

Integrated Air Defense — Golden Dome

Dynamic spectrum reallocation for the nation's integrated missile defense architecture: machine-speed frequency coordination across radars, comms, and EW in the same battlespace. This is the closest existing analog to a real-time spectrum market—automated sensing, AI-driven



Operational deployment, Iraq

decision-making, instant reallocation—at national scale under classified contract.

Homeland Security & Infrastructure Protection

Pursuing DHS Safety Act certification. Pending deployments with the Port Authority of NY/NJ and the Hellenic Ministry of Defense. Demonstrated RF security for airport perimeters and major venue protection ahead of the 2026 FIFA World Cup. Active pipeline for stadium security, power grid RF monitoring, and border surveillance.

GROWTH VECTORS

Spectrum Commoditization

Spectrum access today is binary: license or nothing. Xtremis is building a secondary market where usage rights trade dynamically—priced by location, time, frequency, and interference tolerance. CBRS 3.5 GHz is the near-term proving ground. **Spectrum Exchange** provides sensing and orchestration infrastructure for a spot market via a consortium model uniting holders, financial exchanges, market makers, and enterprise buyers. The endgame: market infrastructure for spectrum as a tradeable commodity, analogous to ICE/CME for energy and derivatives. Dynamic sharing of DoD spectrum alone could unlock trillions in GDP contribution and tens of billions in new government revenue.

Commercial Spectrum Management

Private 5G, ORAN, and unlicensed proliferation are creating RF environments unmanageable with static tools. spectrumOS already solves this for the most demanding user on Earth. Commercial application—interference detection, dynamic channel assignment, QoS assurance—is direct technology transfer. Targets: industrial campuses, logistics hubs, ports, dense urban deployments.

Data-as-a-Service & Proving Ground

Our 1,000-acre **Devil's Den Proving Ground** is the only privately held RF experimentation facility in the U.S. with classified enclave access. Expanding to support DaaS offerings, client experimentation, live demonstrations, and streamlined R&D workflows. **BrightRF** complements live collection with physics-informed synthetic data, enabling ML model training at scale.

Quantum-Enhanced RF Optimization

Spectrum assignment across dozens of systems in a contested EM environment is a combinatorial optimization problem intractable for classical solvers at operational scale. Developing quantum/ML hybrid workflows with academic and industry partners including the Vanderbilt-EPB Institute for Quantum Innovation. Early-stage with significant IP potential at the intersection of quantum computing and national security.

European Expansion

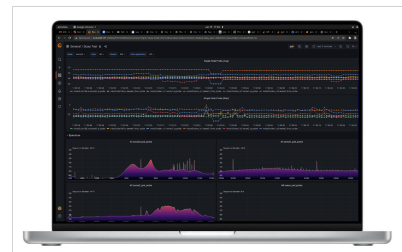
Xtremis operates a subsidiary in Hungary and is expanding presence to Switzerland and Sweden. Selected by the Swedish Ministry of Defence for their contribution to the European Drone Wall initiative. Pursuing partnerships with EPFL and KTH Royal Institute of Technology.



ADSR SP.1 — ground sensor platform



Chameleon — UAS-integrated RF sensing payload



spectrumOS live dashboard



Devil's Den Proving Ground

XTREMIS IN THE NEWS

Oct 2023 **AI-enabled technology developed by Vanderbilt, 101st Airborne deployed for NATO training**

Vanderbilt University News | <https://news.vanderbilt.edu/2023/10/19/ai-enabled-technology-developed-by-vanderbilt-101st-airborne-deployed-for-training-of-nato-and-nato-partner-units-in-europe/>

Apr 2024 **Start-up converting former nuclear reactor site into research facility**

NWA Homepage / KNSA | <https://www.nwahomepage.com/news/start-up-company-converting-former-washington-county-nuclear-reactor-site-into-research-facility/>

Nov 2024 **Xtremis lands over \$80 million in U.S. Army contracts**

Talk Business & Politics | <https://talkbusiness.net/2024/11/xtremis-lands-over-80-million-in-u-s-army-contracts/>

Mar 2025 **Former UA reactor site proves ideal for Xtremis technology development**

Northwest Arkansas Democrat-Gazette | <https://www.nwaonline.com/news/2025/mar/29/former-ua-reactor-site-proves-ideal-for-xtremis/>

THE TEAM

Veteran-owned. 30 personnel across Washington, DC; Fayetteville, AR (Devil's Den Proving Ground); Nashville, TN; and Budapest, Hungary.

Adam Jay Harrison, CEO	Co-Founder. 25+ years in national security and dual-use startups. E&Y Entrepreneur of the Year. Multiple Inc. 500 companies. Distinguished Entrepreneur in Residence and Industry Professor, Vanderbilt University.
Shane Van Wyngaardt, COO	President and COO. Serial entrepreneur with 25+ years in business operations and growth. Specialized in PaaS solutions leveraging AI, data science, and automation for EM spectrum characterization, optimization, and management.
Peter Volgyesi, CTO	Led the Vanderbilt research team (Team Marmot) that won two DARPA Grand Challenges in RF spectrum utilization and wireless network performance. Core technologies from this work form the foundation of the Xtremis IP portfolio.
Brad Treadway, VP Ops	Retired Navy SEAL group commander and intelligence community veteran. 35+ years leading complex technical operations across cyber, electronic warfare, ISR, and high-tech startup environments.
Janos Sallai, VP Eng	20+ years leading R&D and product development in academia and venture-backed companies. Pioneering work in RF localization, angle-of-arrival estimation, and distributed sensing networks.

Partners include Nokia Federal Systems, Booz Allen Hamilton, MIT Lincoln Laboratory, Vanderbilt University, University of Arkansas, and Motorola Solutions. TS/SCI facility clearance (CAGE 93ZJ7).

INVESTMENT

Preparing a \$20M+ Series A to accelerate three priorities:

- **Devil's Den Proving Ground buildout** — DevOps infrastructure for continuous RF experimentation and client demonstration
- **Manufacturing CAPEX** — in-sourcing sensor production to reduce cost and lead time
- **Product development acceleration** — new engineering hires across AI/ML, edge compute, and RF systems

Founded	2021 (Vanderbilt University spinout)
Revenue	\$22.5M booked 2025 \$11.5M gross 2024 net positive
Contracts	\$100M+ DoD portfolio
Deployments	6+ international sites since 2023
Stack	spectrumOS, SpECTRE, VESpER, Spectrum Exchange, BrightRF, ADSR/RogueRF
Facilities	Devil's Den Proving Ground (1,000 ac, AR) classified enclave
Clearances	TS/SCI CAGE 93ZJ7
Team	30 personnel veteran-owned DC, AR, TN, Hungary
Funding	Bootstrapped to date preparing \$20M+ Series A



Make the Invisible Actionable

Real-time spectrum awareness | AI-driven orchestration | Operationalized at scale

We welcome the opportunity to discuss how Xtremis fits within your portfolio or strategic objectives.

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